

WEEK 4: DATA VISUALIZATION & CAPSTONE PROJECT

End-to-End Sales Analysis & Power BI Dashboard

Dataset	Sample Sales Dataset
Records	700
Columns	16
Total Sales	\$118.73M
Total Profit	\$16.89M
Units Sold	1.13M
Visualization Tool	Microsoft Power BI
Analysis Tool	Python / Pandas

This report summarizes the data preparation, Python/Pandas analysis, Power BI visualization, key business findings, and recommendations.

1. Project Objective

The objective of this project is to analyze business sales data, identify important trends and performance drivers, and present the results through an interactive Power BI dashboard for stakeholder decision-making.

2. Dataset Overview

The dataset contains 700 records and 16 columns covering customer segments, countries, products, discounts, units sold, pricing, sales, costs, profit, and date information.

Data quality check	Result
Rows	700
Columns	16
Duplicate rows	0
Missing Discount Band values	53
Other missing values	0

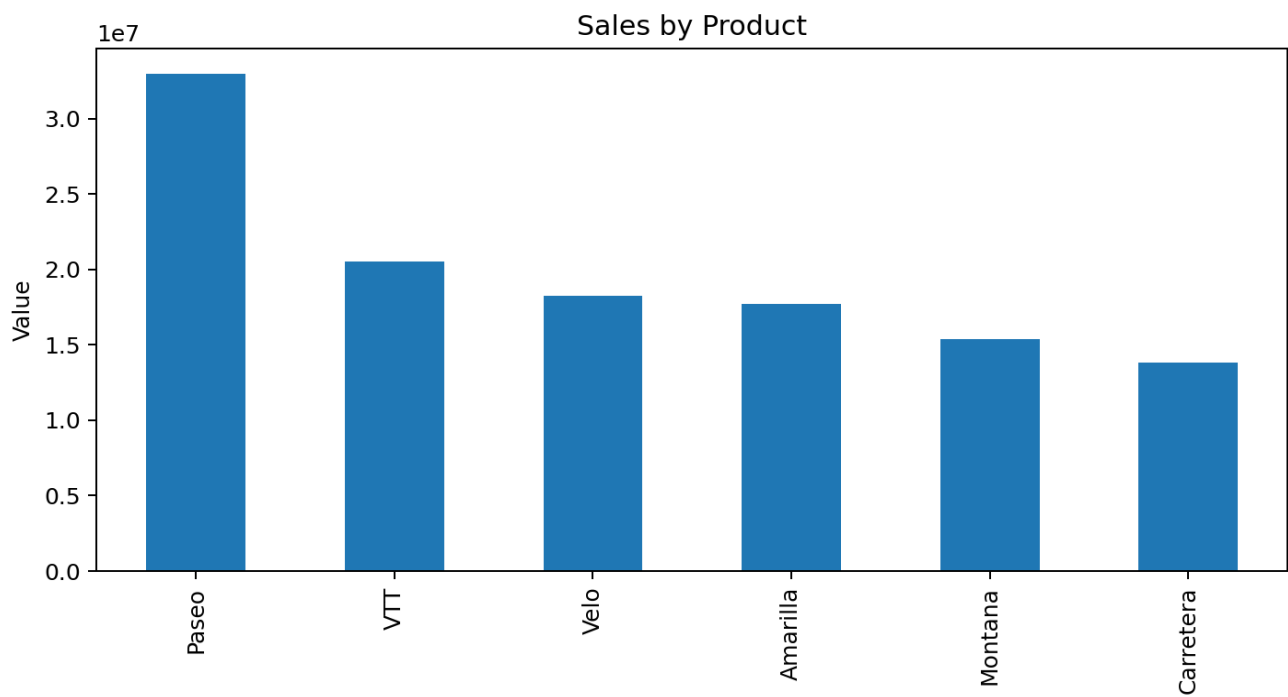
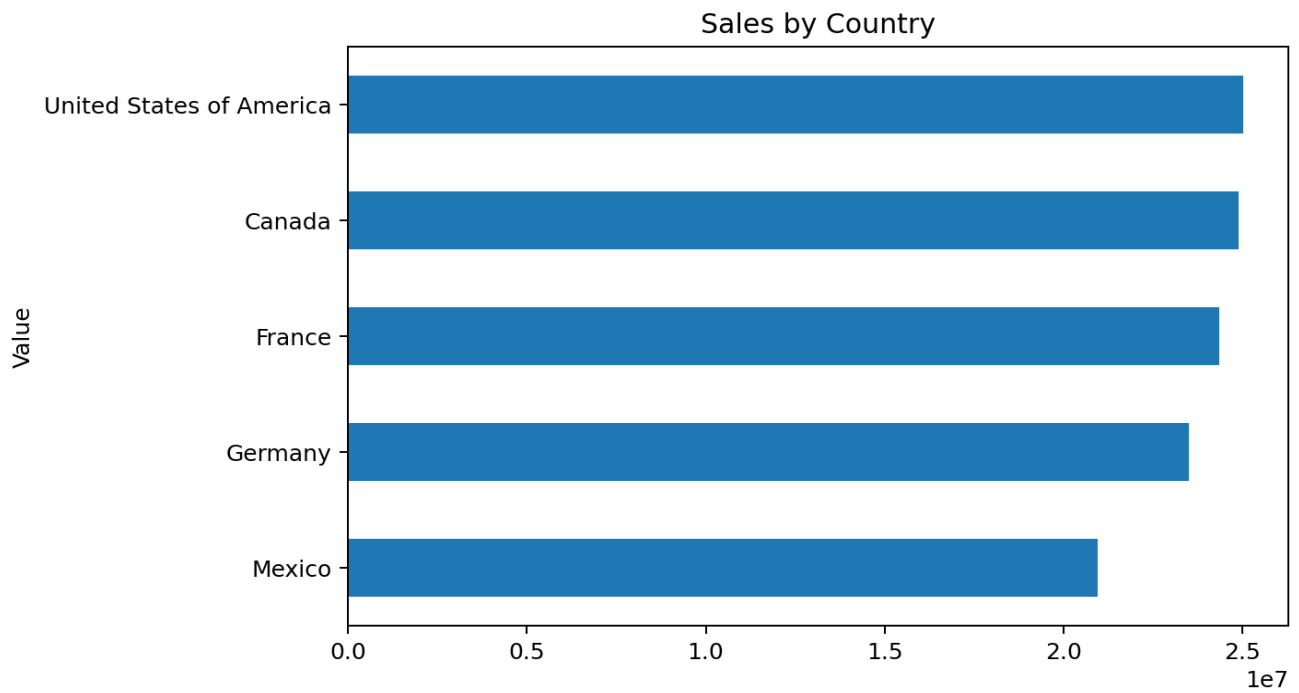
3. Data Cleaning & Python/Pandas Analysis

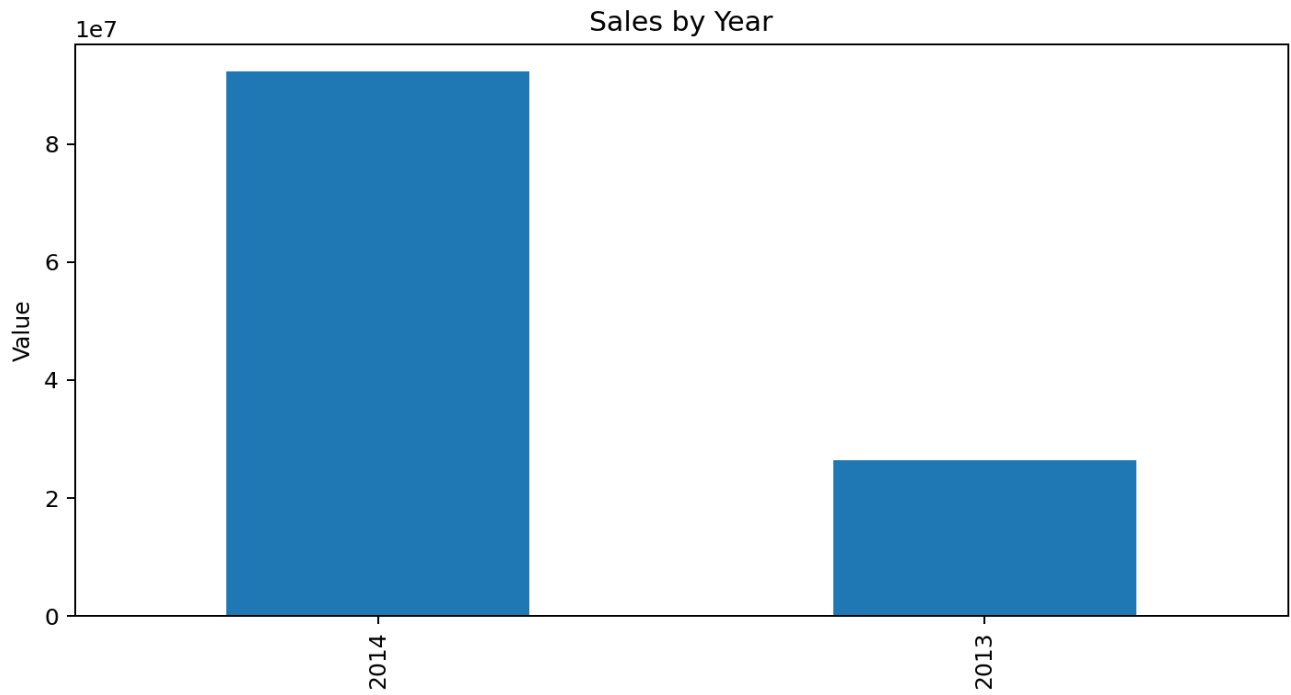
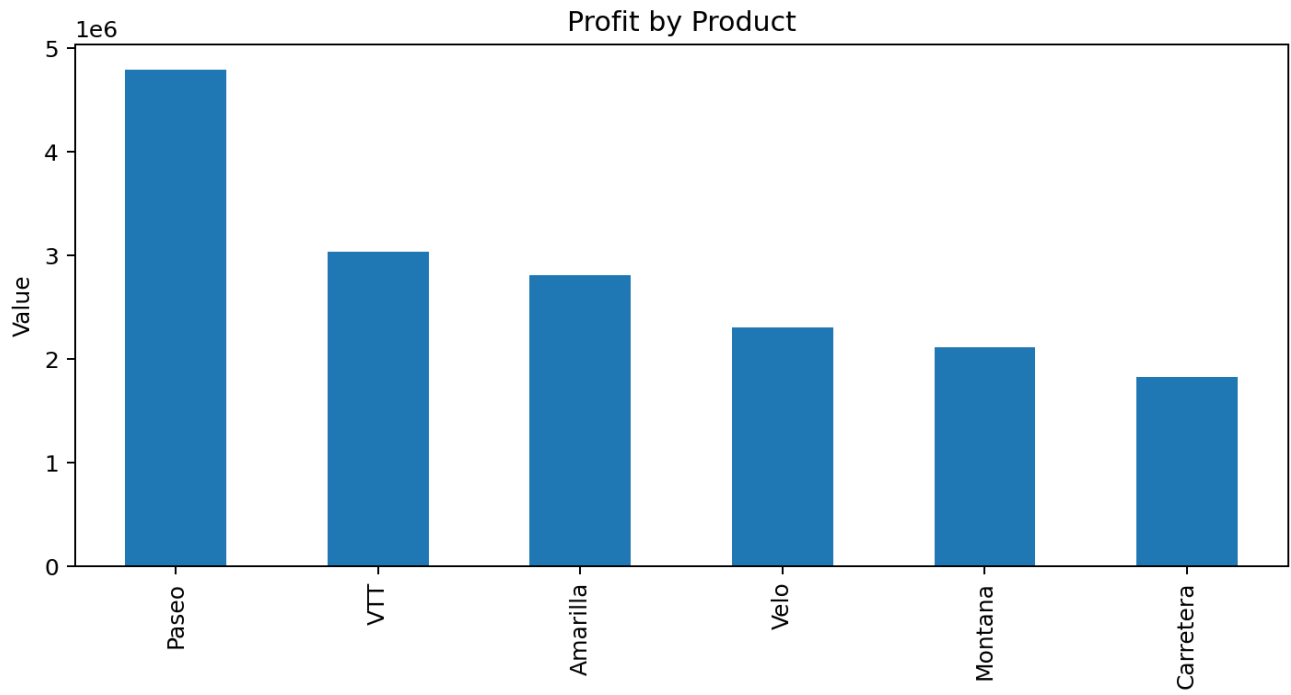
The dataset was loaded into Python using Pandas. Column-name whitespace was removed to ensure consistent field names. Duplicate records were checked and none were found. The 53 missing Discount Band values were handled by assigning the label 'Unknown'. Business metrics and grouped summaries were then calculated by country, product, year, and segment.

4. Key Business Metrics

Metric	Result
Total Sales	\$118.73M
Total Profit	\$16.89M
Total Units Sold	1.13M
Top Country by Sales	United States of America — \$25.03M
Top Product by Sales	Paseo — \$33.01M
Top Product by Profit	Paseo — \$4.80M
Top Segment by Sales	Government — \$52.50M

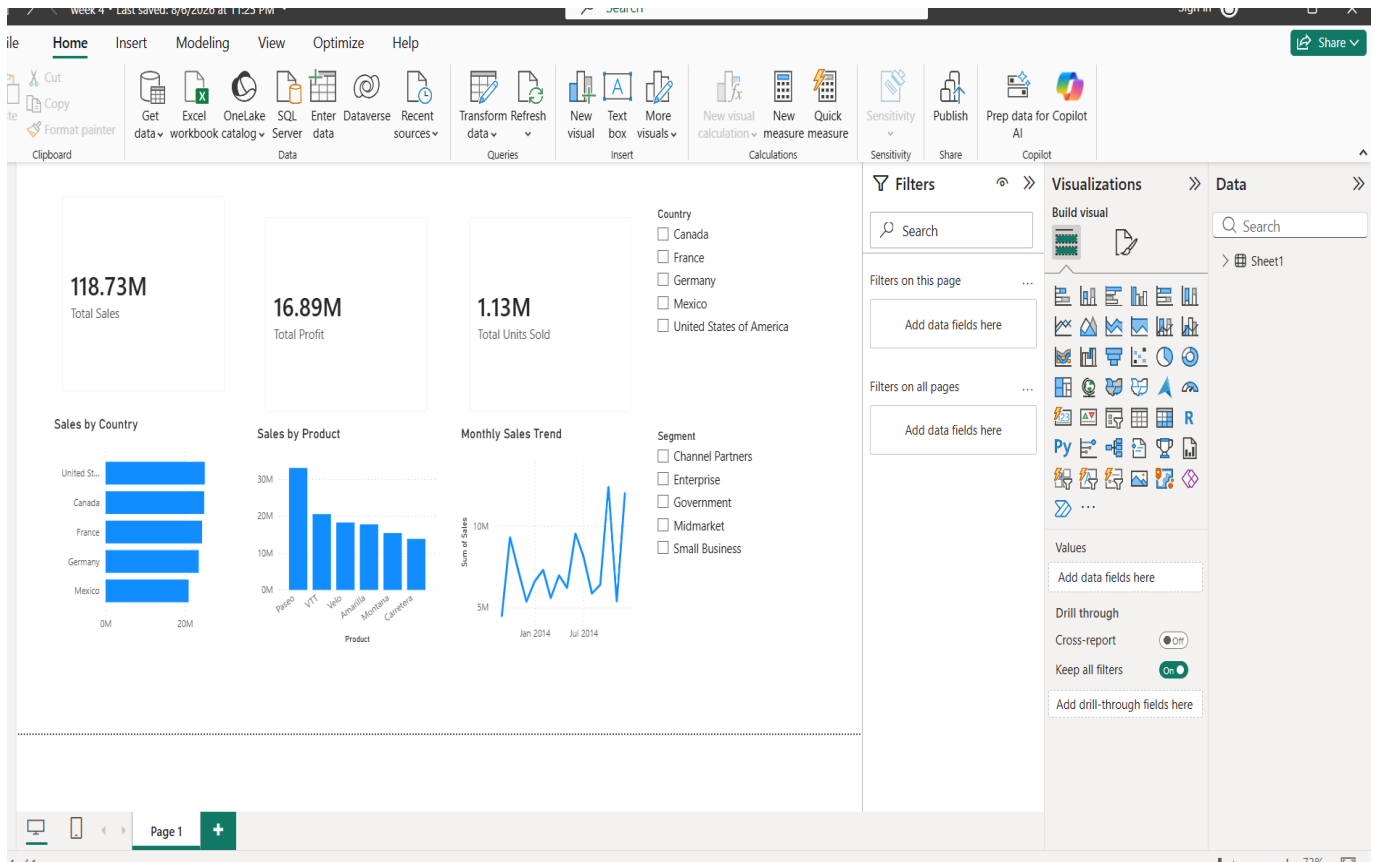
5. Analysis Results





6. Power BI Dashboard

The Power BI dashboard presents KPI cards for Total Sales, Total Profit, and Total Units Sold, along with country, product, and monthly sales visuals. Country and Segment slicers support interactive filtering.



7. Key Insights

1. Total sales reached \$118.73M, with total profit of \$16.89M from approximately 1.13M units sold.
2. United States of America generated the highest country-level sales at \$25.03M.
3. Paseo was the leading product, generating \$33.01M in sales and \$4.80M in profit.
4. Sales in 2014 were \$92.31M, substantially higher than \$26.42M in 2013.
5. The Government segment contributed the highest sales at \$52.50M.

8. Recommendations

1. Continue monitoring and supporting high-performing products such as Paseo while evaluating why other products have lower sales.
2. Investigate the drivers behind the strong performance of the United States and Government segment to identify repeatable growth opportunities.
3. Use the monthly trend in the Power BI dashboard to monitor fluctuations and plan inventory or promotional activity accordingly.
4. Review records with unknown Discount Band values and improve data-entry or source-system controls so future analyses contain fewer missing classifications.
5. Use the dashboard slicers regularly for stakeholder reviews so performance can be compared across countries and customer segments.

9. Conclusion

The project demonstrates an end-to-end analytics workflow: data inspection and cleaning in Python/Pandas, business analysis, and interactive visualization in Power BI. The analysis highlights the major contributors to sales and profit and provides practical directions for further business review.